THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 7: **Particulate Nature of Matter** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A solid has a definite shape because: 1

(a) they have no forces
(b) particles are held in fixed positions by strong forces
(c) they move freely
(d) they are far apart

2. In which situation will a puddle dry the fastest? 1

(a) humid day **(b)** hot and windy day
(c) night **(d)** cold and still day

3. Water vapour in the air condenses on a cold surface. The surface: 1

(a) does not change **(b)** freezes
(c) gets slightly warmed by the vapour **(d)** gets cooler

4. A liquid takes the shape of its container but keeps its volume because: 1

(a) no forces act
(b) particles are far apart
(c) particles slide over each other but stay close
(d) particles are fixed

5. Ice at -10 degrees Celsius is heated steadily. The first change observed is: 1

(a) temperature rises to 0 degrees Celsius **(b)** melting at once
(c) sublimation **(d)** boiling

6. At the melting point of a pure solid, the solid and liquid are in: 1

(a) only liquid form **(b)** only solid form
(c) only gas form **(d)** equilibrium and coexist

7. When ice cubes are placed in a drink, the drink cools because: 1

(a) ice absorbs heat from it to melt **(b)** ice is gas
(c) ice is dry **(d)** ice releases heat

8. A beaker with ice and water at 0 degrees Celsius is heated gently. While some ice remains, the temperature: 1

(a) stays at 0 degrees Celsius **(b)** falls
(c) rises to 100 **(d)** rises slowly

9. Which is NOT true? Particles of a gas: 1

(a) are held in fixed positions **(b)** are widely spaced
(c) collide with walls **(d)** move randomly

10. Which of these shows that particles of matter attract each other? 1

(a) perfume smell spreads
(b) a drop of water holds its shape on a leaf
(c) ink spreads in water
(d) a gas fills a room

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do we use ice to preserve fish? 2

12. A pure solid melts at 80 degrees Celsius. What happens to the temperature while it melts and why? 2

13. Why do we hang wet clothes with a larger spread? 2

14. Why does the level of water in a beaker go down when left uncovered? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why does boiling not occur at 100 degrees Celsius in Shimla, but a higher temperature is reached in a pressure cooker? 3

16. How does the formation of clouds and rain illustrate the changes of state? 3

17. Explain how the water in the oceans can be considered a source of water in the air. 3

OR

Describe an experiment to show that evaporation causes cooling. 3

18. Why does the gas pressure rise when the temperature of a closed container is increased? 3

OR

How does the particle model explain why a gas has low density compared with a solid? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Compare evaporation and boiling in a table of five points and give practical examples. 5

OR

Discuss why climate near the sea is more humid and the effect on comfort and drying of clothes. 5

20. Explain how the three states of matter differ with respect to arrangement, motion, density, compressibility and flow with examples. 5

OR

Explain, with particle diagrams in words, solid, liquid and gas states of a substance and what happens when it is heated, melted and boiled. 5

END OF QUESTION PAPER — Total: 40 Marks